

# BZX84 series

## Voltage regulator diodes

Rev. 7 — 1 January 2023 Product data sheet

## 1. General description

Low-power voltage regulator diodes in a small SOT23 (TO-236AB) Surface-Mounted Device (SMD) plastic package.

The diodes are available in the normalized E24 ±1 % (BZX84-A), ±2 % (BZX84-B) and approximately ±5 % (BZX84-C) tolerance range. The series includes 37 breakdown voltages with nominal working voltages from 2.4 V to 75 V.

## 2. Features and benefits

- Total power dissipation: ≤250 mW
- Three tolerance series: ±1 %, ±2 % and approximately ±5 %
- Working voltage range: nominal 2.4 V to 75 V (E24 range)
- Non-repetitive peak reverse power dissipation: ≤ 40 W

## 3. Applications

- General regulation functions

## 4. Quick reference data

**Table 1. Quick reference data**

| Symbol | Parameter | Conditions | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| $V_F$ | forward voltage | $I_F$ = 10 mA | [1] | - | - | 0.9 | V |
| $P_{tot}$ | total power dissipation | $T_{amb}$ ≤ 25 °C | [2] | - | - | 250 | mW |

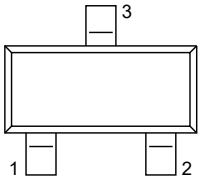
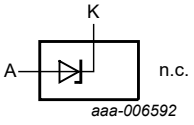
[1] Pulse test: $t_p$ ≤ 100 µs; δ ≤ 0.02.
[2] Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint.

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## 5. Pinning information

**Table 2. Pinning**

| Pin | Symbol | Description | Simplified outline | Graphic symbol |
|---|---|---|---|---|
| 1 | A | anode |  |  |
| 2 | n.c. | not connected | | |
| 3 | K | cathode | | |





## 6. Ordering information

**Table 3. Ordering information**

| Type number | Package | | |
|---|---|---|---|
| | Name | Description | Version |
| BZX84 series[1] | TO-236AB | plastic surface-mounted package; 3 leads | SOT23 |

[1] The series consists of 37 breakdown voltages with nominal working voltages from 2.4 V to 75 V and ±1 %, ±2 % and ±5 % tolerances.


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# 7. Marking

**Table 4. Marking codes**

| Type number | Marking code | Type number | Marking code | Type number | Marking code |
|---|---|---|---|---|---|
| BZX84-A2V4 | %50 | BZX84-B2V4 | %Z0 | BZX84-C2V4 | %T3 |
| BZX84-A2V7 | %51 | BZX84-B2V7 | %Z1 | BZX84-C2V7 | %T4 |
| BZX84-A3V0 | %52 | BZX84-B3V0 | %S1 | BZX84-C3V0 | %T9 |
| BZX84-A3V3 | %53 | BZX84-B3V3 | %S2 | BZX84-C3V3 | %B1 |
| BZX84-A3V6 | %C1 | BZX84-B3V6 | %S3 | BZX84-C3V6 | %B2 |
| BZX84-A3V9 | %55 | BZX84-B3V9 | %S4 | BZX84-C3V9 | %B3 |
| BZX84-A4V3 | %56 | BZX84-B4V3 | %S7 | BZX84-C4V3 | %B6 |
| BZX84-A4V7 | %57 | BZX84-B4V7 | %S8 | BZX84-C4V7 | Z1% |
| BZX84-A5V1 | %58 | BZX84-B5V1 | %R1 | BZX84-C5V1 | Z2% |
| BZX84-A5V6 | %59 | BZX84-B5V6 | %R2 | BZX84-C5V6 | Z3% |
| BZX84-A6V2 | %60 | BZX84-B6V2 | %R5 | BZX84-C6V2 | Z4% |
| BZX84-A6V8 | %61 | BZX84-B6V8 | %R6 | BZX84-C6V8 | Z5% |
| BZX84-A7V5 | %62 | BZX84-B7V5 | %R8 | BZX84-C7V5 | Z6% |
| BZX84-A8V2 | %63 | BZX84-B8V2 | %R9 | BZX84-C8V2 | Z7% |
| BZX84-A9V1 | %64 | BZX84-B9V1 | %T1 | BZX84-C9V1 | Z8% |
| BZX84-A10 | %65 | BZX84-B10 | %66 | BZX84-C10 | Z9% |
| BZX84-A11 | %04 | BZX84-B11 | %Z6 | BZX84-C11 | Y1% |
| BZX84-A12 | %67 | BZX84-B12 | %Z7 | BZX84-C12 | Y2% |
| BZX84-A13 | %C0 | BZX84-B13 | %Z8 | BZX84-C13 | Y3% |
| BZX84-A15 | %69 | BZX84-B15 | %Z9 | BZX84-C15 | Y4% |
| BZX84-A16 | KE% | BZX84-B16 | %70 | BZX84-C16 | Y5% |
| BZX84-A18 | KF% | BZX84-B18 | %71 | BZX84-C18 | Y6% |
| BZX84-A20 | %C2 | BZX84-B20 | %72 | BZX84-C20 | Y7% |
| BZX84-A22 | KG% | BZX84-B22 | %73 | BZX84-C22 | Y8% |
| BZX84-A24 | KH% | BZX84-B24 | %74 | BZX84-C24 | Y9% |
| BZX84-A27 | %75 | BZX84-B27 | %Z5 | BZX84-C27 | %T2 |
| BZX84-A30 | KJ% | BZX84-B30 | %Z4 | BZX84-C30 | %T5 |
| BZX84-A33 | KK% | BZX84-B33 | %Y1 | BZX84-C33 | %T6 |
| BZX84-A36 | %C3 | BZX84-B36 | %Y2 | BZX84-C36 | %T7 |
| BZX84-A39 | %C4 | BZX84-B39 | %S0 | BZX84-C39 | %T8 |
| BZX84-A43 | %C5 | BZX84-B43 | %S5 | BZX84-C43 | %B4 |
| BZX84-A51 | %C6 | BZX84-B47 | %S6 | BZX84-C47 | %B5 |
| BZX84-A75 | %86 | BZX84-B51 | %S9 | BZX84-C51 | %B7 |
| - | - | BZX84-B56 | %R0 | BZX84-C56 | %B8 |
| - | - | BZX84-B62 | %R3 | BZX84-C62 | %B9 |
| - | - | BZX84-B68 | %R4 | BZX84-C68 | %B0 |
| - | - | BZX84-B75 | %R7 | BZX84-C75 | %A1 |


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# 8. Limiting values

**Table 5. Limiting values**

*In accordance with the Absolute Maximum Rating System (IEC 60134).*

| Symbol | Parameter | Conditions | | Min | Max | Unit |
|---|---|---|---|---|---|---|
| $I_F$ | forward current | | | - | 200 | mA |
| $P_{ZSM}$ | non-repetitive peak reverse power dissipation | | [1] | - | 40 | W |
| $P_{tot}$ | total power dissipation | $T_{amb} \leq 25$ °C | [2] | - | 250 | mW |
| $T_j$ | junction temperature | | | - | 150 | °C |
| $T_{amb}$ | ambient temperature | | | -55 | +150 | °C |
| $T_{stg}$ | storage temperature | | | -65 | +150 | °C |

[1] $t_p$ = 100 µs; square wave; $T_j$ = 25 °C prior to surge.
[2] Device mounted on an FR4 PCB, single-sided 70 µm copper, tin-plated and standard footprint.

# 9. Thermal characteristics

**Table 6. Thermal characteristics**

| Symbol | Parameter | Conditions | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| $R_{th(j-a)}$ | thermal resistance from junction to ambient | in free air | [1] | - | - | 500 | K/W |
| $R_{th(j-sp)}$ | thermal resistance from junction to solder point | | [2] | - | - | 330 | K/W |

[1] Device mounted on an FR4 PCB, single-sided 70 µm copper, tin-plated and standard footprint.
[2] Soldering point of cathode tab.

# 10. Characteristics

**Table 7. Characteristics**

*$T_j$ = 25 °C unless otherwise specified.*

| Symbol | Parameter | Conditions | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| $V_F$ | forward voltage | $I_F$ = 10 mA | [1] | - | - | 0.9 | V |

[1] Pulse test: $t_p \leq 100$ µs; $\delta \leq 0.02$


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**Table 8. Characteristics per type; BZX84-A2V4 to BZX84-C24**

*$T_j$ = 25 °C unless otherwise specified.*

| BZX84 -xxx | Sel | Working voltage $V_Z$ (V) $I_Z$ = 5 mA | | Maximum differential resistance $r_{dif}$ (Ω) | | Reverse current $I_R$ (μA) | | Temperature coefficient $S_Z$ (mV/K) $I_Z$ = 5 mA | | Diode capacitance $C_d$ (pF) [1] | Non-repetitive peak reverse current $I_{ZSM}$ (A) [2] |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Min | Max | $I_Z$ = 1 mA | $I_Z$ = 5 mA | Max | $V_R$ (V) | Min | Max | Max | Max |
| 2V4 | A | 2.37 | 2.43 | 600 | 100 | 50 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 2.35 | 2.45 | | | | | | | | |
| | C | 2.20 | 2.60 | | | | | | | | |
| 2V7 | A | 2.67 | 2.73 | 600 | 100 | 20 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 2.65 | 2.75 | | | | | | | | |
| | C | 2.50 | 2.90 | | | | | | | | |
| 3V0 | A | 2.97 | 3.03 | 600 | 95 | 10 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 2.94 | 3.06 | | | | | | | | |
| | C | 2.80 | 3.20 | | | | | | | | |
| 3V3 | A | 3.26 | 3.34 | 600 | 95 | 5 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 3.23 | 3.37 | | | | | | | | |
| | C | 3.10 | 3.50 | | | | | | | | |
| 3V6 | A | 3.56 | 3.64 | 600 | 90 | 5 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 3.53 | 3.67 | | | | | | | | |
| | C | 3.40 | 3.80 | | | | | | | | |
| 3V9 | A | 3.86 | 3.94 | 600 | 90 | 3 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 3.82 | 3.98 | | | | | | | | |
| | C | 3.70 | 4.10 | | | | | | | | |
| 4V3 | A | 4.25 | 4.35 | 600 | 90 | 3 | 1 | -3.5 | 0.0 | 450 | 6.0 |
| | B | 4.21 | 4.39 | | | | | | | | |
| | C | 4.00 | 4.60 | | | | | | | | |
| 4V7 | A | 4.65 | 4.75 | 500 | 80 | 3 | 2 | -3.5 | 0.2 | 300 | 6.0 |
| | B | 4.61 | 4.79 | | | | | | | | |
| | C | 4.40 | 5.00 | | | | | | | | |
| 5V1 | A | 5.04 | 5.16 | 480 | 60 | 2 | 2 | -2.7 | 1.2 | 300 | 6.0 |
| | B | 5.00 | 5.20 | | | | | | | | |
| | C | 4.80 | 5.40 | | | | | | | | |
| 5V6 | A | 5.54 | 5.66 | 400 | 40 | 1 | 2 | -2.0 | 2.5 | 300 | 6.0 |
| | B | 5.49 | 5.71 | | | | | | | | |
| | C | 5.20 | 6.00 | | | | | | | | |
| 6V2 | A | 6.13 | 6.27 | 150 | 10 | 3 | 4 | 0.4 | 3.7 | 200 | 6.0 |
| | B | 6.08 | 6.32 | | | | | | | | |
| | C | 5.80 | 6.60 | | | | | | | | |
| 6V8 | A | 6.73 | 6.87 | 80 | 15 | 2 | 4 | 1.2 | 4.5 | 200 | 6.0 |
| | B | 6.66 | 6.94 | | | | | | | | |
| | C | 6.40 | 7.20 | | | | | | | | |
| 7V5 | A | 7.42 | 7.58 | 80 | 15 | 1 | 5 | 2.5 | 5.3 | 150 | 4.0 |
| | B | 7.35 | 7.65 | | | | | | | | |
| | C | 7.00 | 7.90 | | | | | | | | |


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| BZX84 -xxx | Sel | Working voltage $V_Z$ (V) $I_Z$ = 5 mA | | Maximum differential resistance $r_{dif}$ (Ω) | | Reverse current $I_R$ (μA) | | Temperature coefficient $S_Z$ (mV/K) $I_Z$ = 5 mA | | Diode capacitance $C_d$ (pF) [1] | Non-repetitive peak reverse current $I_{ZSM}$ (A) [2] |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Min | Max | $I_Z$ = 1 mA | $I_Z$ = 5 mA | Max | $V_R$ (V) | Min | Max | Max | Max |
| 8V2 | A | 8.11 | 8.29 | 80 | 15 | 0.7 | 5 | 3.2 | 6.2 | 150 | 4.0 |
| | B | 8.04 | 8.36 | | | | | | | | |
| | C | 7.70 | 8.70 | | | | | | | | |
| 9V1 | A | 9.00 | 9.20 | 100 | 15 | 0.5 | 6 | 3.8 | 7.0 | 150 | 3.0 |
| | B | 8.92 | 9.28 | | | | | | | | |
| | C | 8.50 | 9.60 | | | | | | | | |
| 10 | A | 9.90 | 10.10 | 150 | 20 | 0.2 | 7 | 4.5 | 8.0 | 90 | 3.0 |
| | B | 9.80 | 10.20 | | | | | | | | |
| | C | 9.40 | 10.60 | | | | | | | | |
| 11 | A | 10.89 | 11.11 | 150 | 20 | 0.1 | 8 | 5.4 | 9.0 | 85 | 2.5 |
| | B | 10.80 | 11.20 | | | | | | | | |
| | C | 10.40 | 11.60 | | | | | | | | |
| 12 | A | 11.88 | 12.12 | 150 | 25 | 0.1 | 8 | 6.0 | 10.0 | 85 | 2.5 |
| | B | 11.80 | 12.20 | | | | | | | | |
| | C | 11.40 | 12.70 | | | | | | | | |
| 13 | A | 12.87 | 13.13 | 170 | 30 | 0.1 | 8 | 7.0 | 11.0 | 80 | 2.5 |
| | B | 12.70 | 13.30 | | | | | | | | |
| | C | 12.40 | 14.10 | | | | | | | | |
| 15 | A | 14.85 | 15.15 | 200 | 30 | 0.05 | 10.5 | 9.2 | 13.0 | 75 | 2.0 |
| | B | 14.70 | 15.30 | | | | | | | | |
| | C | 13.80 | 15.60 | | | | | | | | |
| 16 | A | 15.84 | 16.16 | 200 | 40 | 0.05 | 11.2 | 10.4 | 14.0 | 75 | 1.5 |
| | B | 15.70 | 16.30 | | | | | | | | |
| | C | 15.30 | 17.10 | | | | | | | | |
| 18 | A | 17.82 | 18.18 | 225 | 45 | 0.05 | 12.6 | 12.4 | 16.0 | 70 | 1.5 |
| | B | 17.60 | 18.40 | | | | | | | | |
| | C | 16.80 | 19.10 | | | | | | | | |
| 20 | A | 19.80 | 20.20 | 225 | 55 | 0.05 | 14 | 14.4 | 18.0 | 60 | 1.5 |
| | B | 19.60 | 20.40 | | | | | | | | |
| | C | 18.80 | 21.20 | | | | | | | | |
| 22 | A | 21.78 | 22.22 | 250 | 55 | 0.05 | 15.4 | 16.4 | 20.0 | 60 | 1.25 |
| | B | 21.60 | 22.40 | | | | | | | | |
| | C | 20.80 | 23.30 | | | | | | | | |
| 24 | A | 23.76 | 24.24 | 250 | 70 | 0.05 | 16.8 | 18.4 | 22.0 | 55 | 1.25 |
| | B | 23.50 | 24.50 | | | | | | | | |
| | C | 22.80 | 25.60 | | | | | | | | |

[1] f = 1 MHz; $V_R$ = 0 V
[2] $t_p$ = 100 μs; square wave; $T_j$ = 25 °C


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**Table 9. Characteristics per type; BZX84-A27 to BZX84-C75**

*$T_j$ = 25 °C unless otherwise specified.*

| BZX84 -xxx | Sel | Working voltage $V_Z$ (V) $I_Z$ = 2 mA | | Maximum differential resistance $r_{dif}$ (Ω) | | Reverse current $I_R$ (μA) | | Temperature coefficient $S_Z$ (mV/K) $I_Z$ = 2 mA | | Diode capacitance $C_d$ (pF) [1] | Non-repetitive peak reverse current $I_{ZSM}$ (A) [2] |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Min | Max | $I_Z$ = 0.5 mA | $I_Z$ = 2 mA | Max | $V_R$ (V) | Min | Max | Max | Max |
| 27 | A | 26.73 | 27.27 | 300 | 80 | 0.05 | 18.9 | 21.4 | 25.3 | 50 | 1.0 |
| | B | 26.50 | 27.50 | | | | | | | | |
| | C | 25.10 | 28.90 | | | | | | | | |
| 30 | A | 29.70 | 30.30 | 300 | 80 | 0.05 | 21 | 24.4 | 29.4 | 50 | 1.0 |
| | B | 29.40 | 30.60 | | | | | | | | |
| | C | 28.00 | 32.00 | | | | | | | | |
| 33 | A | 32.67 | 33.33 | 325 | 80 | 0.05 | 23.1 | 27.4 | 33.4 | 45 | 0.9 |
| | B | 32.30 | 33.70 | | | | | | | | |
| | C | 31.00 | 35.00 | | | | | | | | |
| 36 | A | 35.64 | 36.36 | 350 | 90 | 0.05 | 25.2 | 30.4 | 37.4 | 45 | 0.8 |
| | B | 35.30 | 36.70 | | | | | | | | |
| | C | 34.00 | 38.00 | | | | | | | | |
| 39 | A | 38.61 | 39.39 | 350 | 130 | 0.05 | 27.3 | 33.4 | 41.2 | 45 | 0.7 |
| | B | 38.20 | 39.80 | | | | | | | | |
| | C | 37.00 | 41.00 | | | | | | | | |
| 43 | A | 42.57 | 43.43 | 375 | 150 | 0.05 | 30.1 | 37.6 | 46.6 | 40 | 0.6 |
| | B | 42.10 | 43.90 | | | | | | | | |
| | C | 40.00 | 46.00 | | | | | | | | |
| 47 | B | 46.10 | 47.90 | 375 | 170 | 0.05 | 32.9 | 42.0 | 51.8 | 40 | 0.5 |
| | C | 44.00 | 50.00 | | | | | | | | |
| 51 | A | 50.49 | 51.51 | 400 | 180 | 0.05 | 35.7 | 46.6 | 57.2 | 40 | 0.4 |
| | B | 50.00 | 52.00 | | | | | | | | |
| | C | 48.00 | 54.00 | | | | | | | | |
| 56 | B | 54.90 | 57.10 | 425 | 200 | 0.05 | 39.2 | 52.2 | 63.8 | 40 | 0.3 |
| | C | 52.00 | 60.00 | | | | | | | | |
| 62 | B | 60.80 | 63.20 | 450 | 215 | 0.05 | 43.4 | 58.8 | 71.6 | 35 | 0.3 |
| | C | 58.00 | 66.00 | | | | | | | | |
| 68 | B | 66.60 | 69.40 | 475 | 240 | 0.05 | 47.6 | 65.6 | 79.8 | 35 | 0.25 |
| | C | 64.00 | 72.00 | | | | | | | | |
| 75 | A | 74.25 | 75.75 | 500 | 255 | 0.05 | 52.5 | 73.4 | 88.6 | 35 | 0.20 |
| | B | 73.50 | 76.50 | | | | | | | | |
| | C | 70.00 | 79.00 | | | | | | | | |

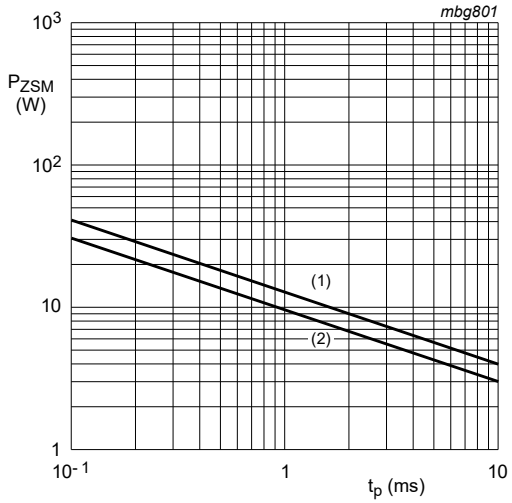
[1] f = 1 MHz; $V_R$ = 0 V

[2] $t_p$ = 100 μs; square wave; $T_j$ = 25 °C


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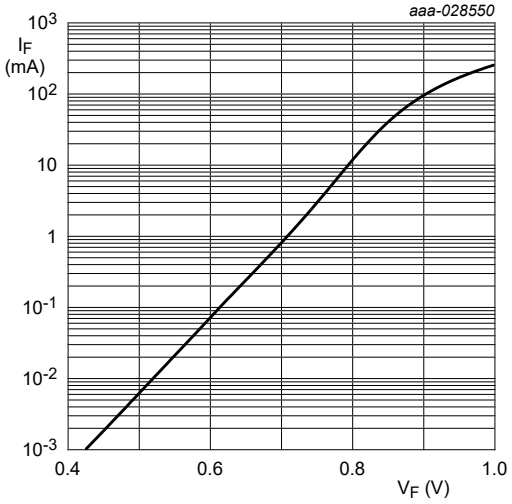

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(1) $T_j$ = 25 °C (before surge)

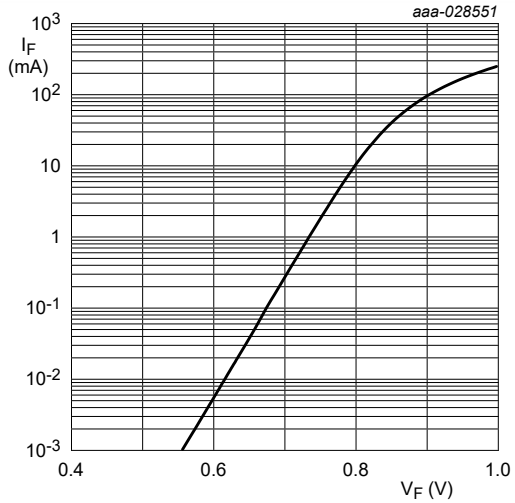
(2) $T_j$ = 150 °C (before surge)

**Fig. 1. Non-repetitive peak reverse power dissipation as a function of pulse duration; maximum values**



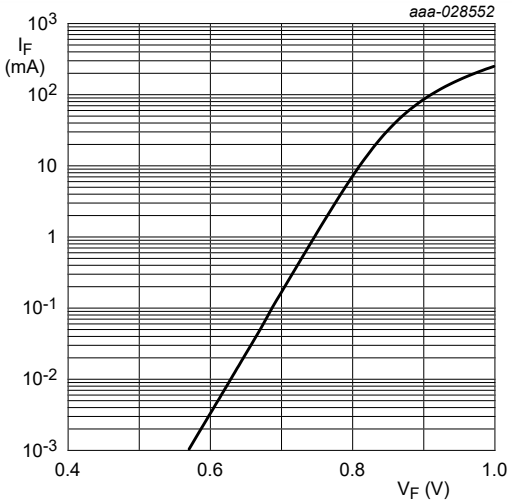

$T_j$ = 25 °C

**Fig. 2. Forward current as a function of forward voltage; typical values (BZX84-A/B/C2V4)**




$T_j$ = 25 °C

**Fig. 3. Forward current as a function of forward voltage; typical values (BZX84-A/B/C6V8)**



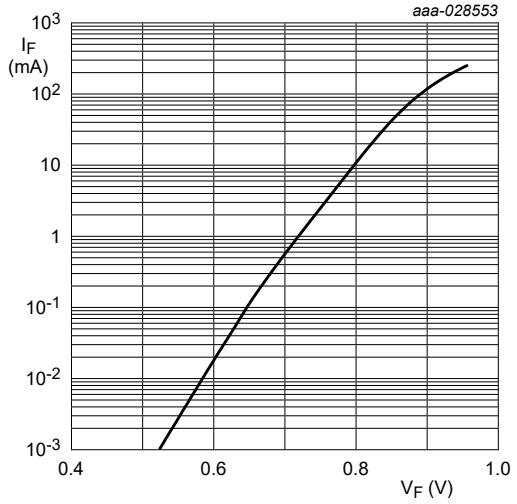

$T_j$ = 25 °C

**Fig. 4. Forward current as a function of forward voltage; typical values (BZX84-A/B/C7V5)**


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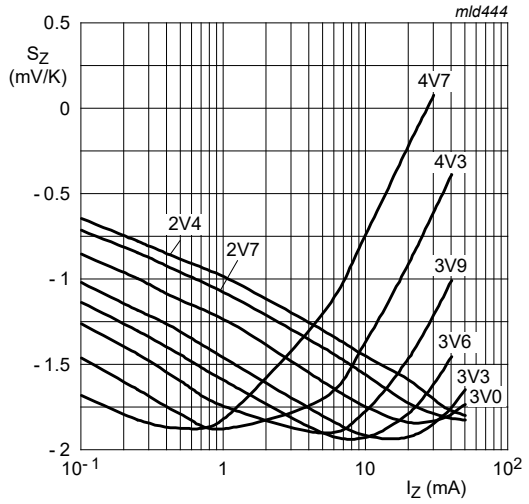
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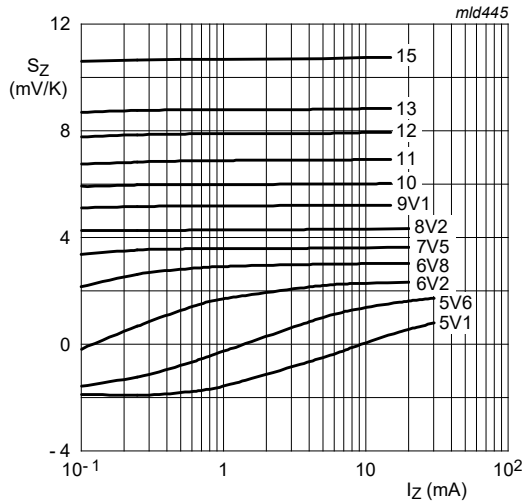

$T_j$ = 25 °C

**Fig. 5. Forward current as a function of forward voltage; typical values (BZX84-A/B/C75)**



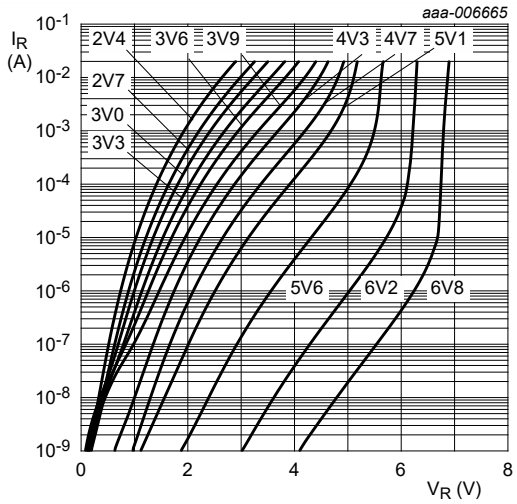

$T_j$ = 25 °C to 150 °C

**Fig. 6. Temperature coefficient as a function of working current; typical values (BZX84-A/B/C2V4 to BZX84-A/B/C4V7)**




$T_j$ = 25 °C to 150 °C

**Fig. 7. Temperature coefficient as a function of working current; typical values (BZX84-A/B/C5V1 to BZX84-A/B/C15)**



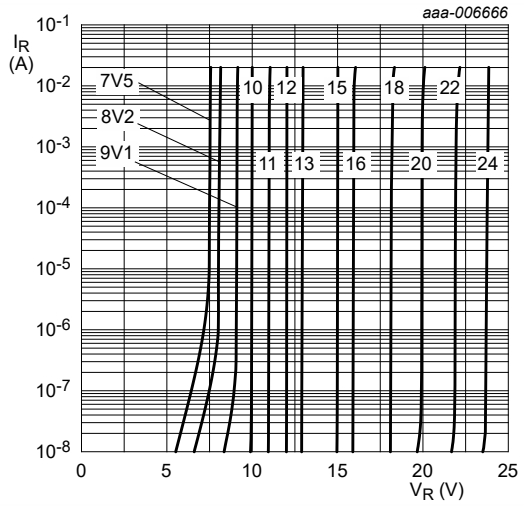

$T_j$ = 25 °C

**Fig. 8. Reverse current as a function of reverse voltage; typical values (BZX84-A/B/C2V4 to BZX84-A/B/C6V8)**


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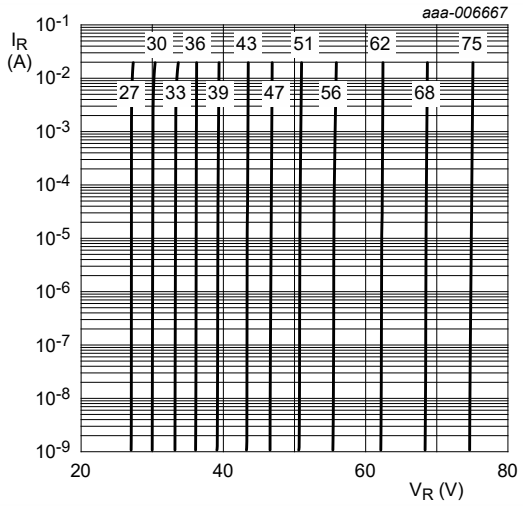
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$T_j$ = 25 °C

**Fig. 9. Reverse current as a function of reverse voltage; typical values (BZX84-A/B/C7V5 to BZX84-A/B/C24)**




$T_j$ = 25 °C

**Fig. 10. Reverse current as a function of reverse voltage; typical values (BZX84-A/B/C27 to BZX84-A/B/C75)**

## 11. Package outline

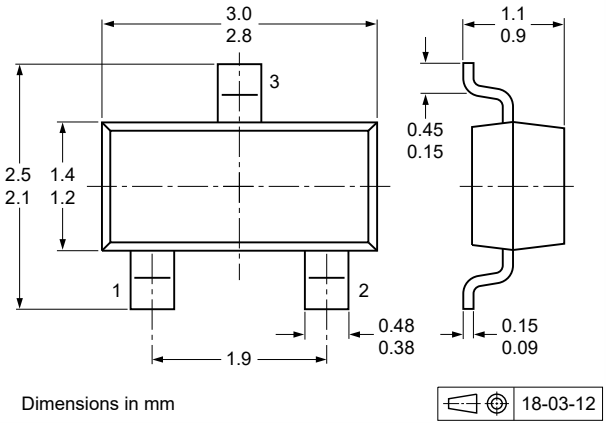



**Fig. 11. Package outline SOT23 (TO-236AB)**


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## 12. Soldering

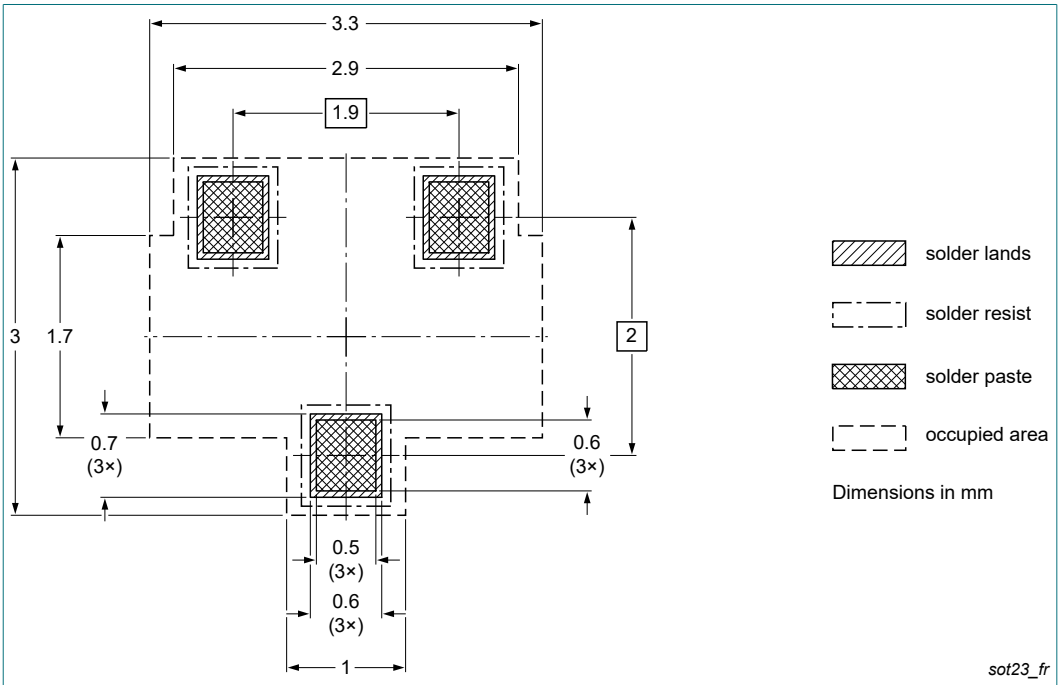



**Fig. 12. Reflow soldering footprint for SOT23 (TO-236AB)**

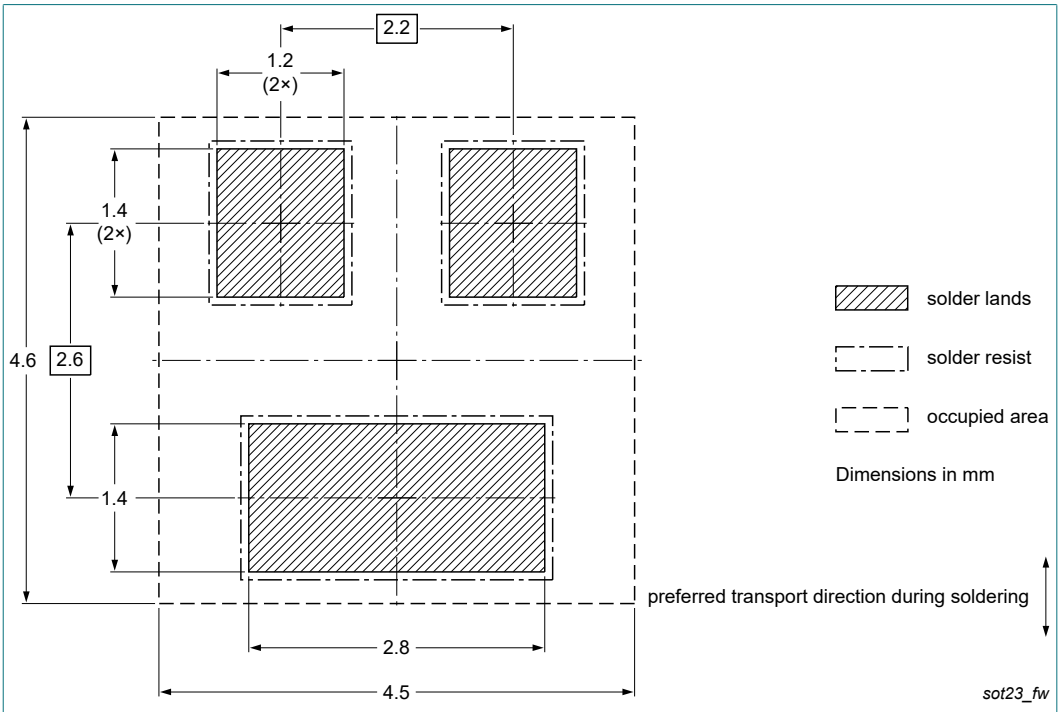



**Fig. 13. Wave soldering footprint for SOT23 (TO-236AB)**


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# 13. Revision history

**Table 10. Revision history**

| Document ID | Release date | Data sheet status | Change notice | Supersedes |
|---|---|---|---|---|
| BZX84_SER v.7 | 20230101 | Product data sheet | - | BZX84_SER v.6 |
| Modifications: | • Section "Packing information" removed<br>• Limiting values: Temperature specifications adjusted<br>• Products changed to non-automotive qualification. Please refer to nexperia.com for automotive (-Q) product alternatives. | | | |
| BZX84_SER v.6 | 20140306 | Product data sheet | | BZX84_SER v.5 |
| BZX84_SER v.5 | 20130918 | Product data sheet | - | BZX84_SER v.4 |
| BZX84_SER v.4 | 20130322 | Product data sheet | - | BZX84_SERIES v.3 |
| BZX84_SERIES v.3 | 20030410 | Product data sheet | - | BZX84 v.2 |
| BZX84 v.2 | 19990518 | Product specification | - | BZX84 v.1 |
| BZX84 v.1 | 19960426 | Product specification | - | - |


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# 14. Legal information

## Data sheet status

| Document status [1][2] | Product status [3] | Definition |
|---|---|---|
| Objective [short] data sheet | Development | This document contains data from the objective specification for product development. |
| Preliminary [short] data sheet | Qualification | This document contains data from the preliminary specification. |
| Product [short] data sheet | Production | This document contains the product specification. |

[1] Please consult the most recently issued document before initiating or completing a design.
[2] The term 'short data sheet' is explained in section "Definitions".
[3] The product status of device(s) described in this document may have changed since this document was published and may differ in case of multiple devices. The latest product status information is available on the internet at https://www.nexperia.com.

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